

WIRELESS NETWORKING

Day – 2 TCP/IP Suite Theory

Task – 1 Wireshark and identify the packet

- I cannot directly run applications on your computer or interact with network interfaces to capture live traffic. However, I can guide you through performing this analysis yourself step-by-step using Wireshark.
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- **Step-by-Step Instructions**
- Launch Wireshark and select your active network interface (e.g., *Wi-Fi* or *Ethernet*).
- Click the Start Capture button (the blue shark fin icon).
- Open your web browser and navigate to (or hard refresh using Ctrl + F5 / Cmd + Shift + R).
- Switch back to Wireshark and click the Stop Capture button (the red square icon).
- In the filter bar at the top, apply a filter to find traffic related to your browser session (e.g., `tls or dns or ip.addr == <Flipkart IP>`).
- **Layer-by-Layer Packet & Header Breakdown**
- Select any HTTP/TLS packet sent during the request to Flipkart. In Wireshark's Packet Details (middle pane), expand each section to inspect the headers for all four layers of the TCP/IP model:
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- **1. Application Layer**
- Protocol Example: HTTP, HTTP/2, HTTP/3, or TLS (Transport Layer Security) / DNS.
- If using HTTP/1.1 or HTTP/2, look for fields such as Host: [, User-Agent, GET / HTTP/1.1, or Server Name Indication (SNI): inside the TLS Handshake Server Name extension.
- Function: Provides network services directly to end-user applications (e.g., requesting web pages via HTTPS).
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- *2. Transport Layer*
- Protocol Example: TCP (Transmission Control Protocol) or UDP (if using HTTP/3 / QUIC).
- Source Port: A ephemeral port assigned by your OS (e.g., 54321).
- Destination Port: 443 (standard port for HTTPS).
- Sequence Number & Acknowledgment Number: Used for reliable delivery and flow control.
- Function: Manages end-to-end communication, error recovery, and data segmentation between the client and server.
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- *3. Internet Layer (Network Layer)*
- Protocol Example: IPv4 or IPv6.
- Source IP Address: Your computer's local IP address (e.g., 192.168.1.X).
- Destination IP Address: Flipkart's server IP address (e.g., resolved via DNS).
- Time to Live (TTL): Prevents packets from looping endlessly in the network.
- Function: Handles logical addressing and routing of packets across network boundaries.
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- *4. Network Access Layer (Link / Data Link Layer)*
- Protocol Example: Ethernet II or IEEE 802.11 (Wi-Fi).
- Source MAC Address: Your physical Network Interface Card (NIC) MAC address.
- Destination MAC Address: Your local router/gateway's MAC address.
- EtherType: Indicates the encapsulated protocol (e.g., IPv4 (0x0800)).
- Function: Handles physical addressing and frame transmission over the local physical medium (cable or wireless).
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Task – 2 OSI Layers and TCP Layer.

OSI Layer	OSI Layer Name	TCP/IP Layer Name	TCP/IP Layer	Real-World App Example & Description
7	Application	Application	4	WhatsApp / Zomato Interface: You tap "Send" on WhatsApp or tap "Place Order" on

				Zomato. Protocols like HTTPS or MQTT/WebSockets format and transmit the data.
6	Presentation	(Combined in Application)	4	Encryption & Formatting: WhatsApp encrypts your text using Signal Protocol (End-to-End Encryption) and formats text/images (JSON, JPEG). Zomato encrypts payment details via TLS/SSL.
5	Session	(Combined in Application)	4	Session Management: Keeps your Zomato checkout session or WhatsApp active connection open so the server knows it's still you without asking you to log in repeatedly.
4	Transport	Transport	3	TCP / UDP Delivery: • TCP (Zomato order / WhatsApp text): Ensures 100% reliable, ordered delivery via acknowledgments. • UDP (WhatsApp Voice/Video Call): Prioritizes real-time speed over retransmitting lost packets.
3	Network	Internet	2	IP Routing: Your phone's IP address package gets routed

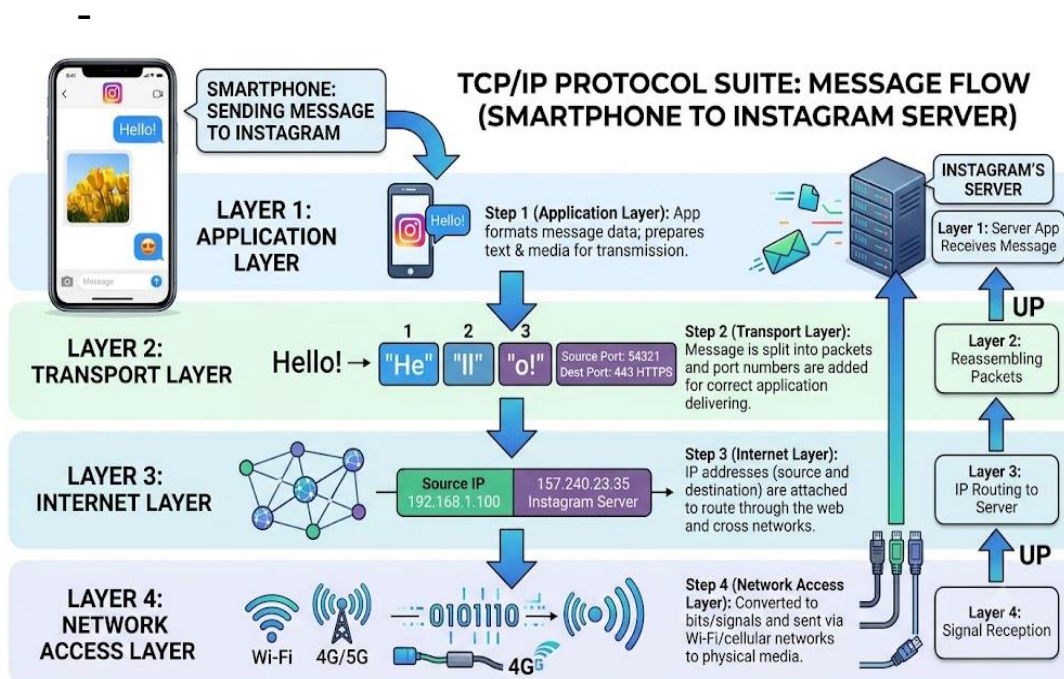
				across the internet (routers, ISP) to reach Zomato's or WhatsApp's server IP address using IPv4/IPv6.
2	Data Link	Network Access (or Link)	1	Local Frame Delivery: Data is packaged into Ethernet/Wi-Fi frames using physical MAC addresses to travel from your phone to your home Wi-Fi router or cellular tower.
1	Physical	Network Access (or Link)	1	Hardware Transmission: Radio frequency signals over Wi-Fi / 5G / 4G or electrical/optical pulses over fiber cables carrying binary data (0s and 1s).

Task – 3 Transport layer and it works.

- Source Port (16 bits): Identifies the specific application or process sending the data on the source host.
- Destination Port (16 bits): Identifies the target application or process that should receive the data on the destination host.
- Sequence Number (32 bits): Tracks the order of transmitted bytes to ensure reliable, in-sequence reassembly of data streams at the destination.
- Acknowledgment Number (32 bits): Indicates the next expected byte sequence number from the sender, confirming receipt of previous bytes.
- Data Offset / Header Length (4 bits): Specifies the size of the TCP header in 32-bit words, indicating where the payload (data) begins.
- Reserved (3 or 6 bits): Set aside for future use or experimental features (typically set to zero).

- Control Flags / Control Bits (9 bits total, e.g., URG, ACK, PSH, RST, SYN, FIN): Control bits used to manage connection establishment, termination, data transfer behavior, and state control.
- Window Size (16 bits): Specifies the number of bytes the receiver is currently willing to accept, facilitating flow control.
- Checksum (16 bits): Provides error-checking capabilities for both the TCP header and data to detect corruption during transit.
- Urgent Pointer (16 bits): Indicates the offset to urgent data within the payload when the URG flag is set.
- Options (0 to 320 bits, variable): Optional settings (such as Maximum Segment Size or Window Scale) used to negotiate TCP capabilities.

Task – 4 How msg sends to Instagram server.



Task – 5 Captured in Wireshark.

- Selected Packet Example
- Protocol: HTTP / TCP
- Source Info: 192.168.1.15:52410 → 104.18.2.35:80
- Packet Detail: POST /api/v1/book-ticket HTTP/1.1

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- **1. TCP/IP Layer**
- Layer: Application Layer (Layer 5 in the modern 5-layer TCP/IP model, or Layer 7 in the OSI model).
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- **2. How It Was Identified**
- Wireshark explicitly decodes the top-level payload as HTTP (Hypertext Transfer Protocol).
- Packet Dissection Panel:
- Under the Hypertext Transfer Protocol section, the request method is listed as POST, and the request URI is /api/v1/book-ticket.
- The transport layer beneath it is TCP (Transmission Control Protocol) targeting destination port 80 (standard HTTP port).
- The payload content contains JSON data formatted like {"event_id": "EV12345", "seats": ["A1", "A2"]}
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- **3. Real-World Action Represented**
- App Action: Booking a movie or event ticket on BookMyShow (or a similar ticketing app).
- When you select your seats and tap "Proceed to Pay" or "Confirm Booking" inside the app, the application constructs a payload containing your selected event details and user details. It then sends an HTTP POST request to the backend servers to initiate the reservation transaction.
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